

Study of the convergence parameters

In order to plot the HHG-C spectra, we currently rely on different formula. One of them reads

$$Q(\omega) = \left| -\dot{d}(\omega) \right|^2, \quad \text{Eq. (22)} \quad (1)$$

where the dipole moment $d(t)$ is given by

$$d(t) = - \int \psi^*(x, t) \hat{x} \psi(x, t) dx, \quad (2)$$

and its Fourier transform, by the relation

$$d(\omega) = \int e^{i\omega t} d(t) dt. \quad (3)$$

In order to deal with the derivative in Eq. (2), we currently use the following formula

$$f'(x_i) = \frac{f'_l(x_i) + f'_r(x_i)}{2} \quad (4)$$

with

$$f'_l(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}, \quad f'_r(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}}.$$

Let us note that a dot express differentiation in time. In this document we want to evaluate how accurate is the approximation expressed by Eq. (4).

1 Test on usual, continuous and differentiable function

In this section, we want to see how accurately Eq. (4) approximates the derivative of a smooth and differentiable function. The function we are using for this test is $f(x) = \sin(x)$. So $f'(x) = \cos(x)$. Fig. (??) shows the relative error between the analytical formula and the numerical estimation, according to the formula

$$\sigma_{\text{rel}} = \left| \frac{f'(x) - \cos(x)}{\cos(x)} \right|. \quad (5)$$

Let us recall that $f(x) = \sin(x)$ here. The precision depends on the number of steps

$$\Delta x = (b - a)/N, \quad b = -a = \pi. \quad (6)$$

As shown in the figure, the approximation expressed in Eq. (4) is of order 2, except on the boundaries. The following table gives the numerical values for σ_{rel}

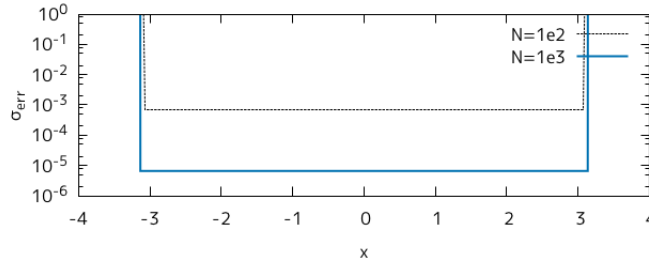


Figure 1: Relative error to the reference for Eq. (4)

$N = 100$			$N = 1000$		
i	x_i	σ_{rel}	i	x_i	σ_{rel}
1	-3.14159265358979	259034758306034.	1	-3.14159265358979	25678644868791.8
2	-3.07812613533545	6.711979630687218(-4)	2	-3.13530317880783	6.592902546370755E(-6)
⋮	⋮	⋮	⋮	⋮	⋮
49	-9.519977738150875(-2)	6.711979630665886(-4)	499	-9.434212172942533(-3)	6.592901664379963(-6)
50	-3.173325912716950(-2)	6.711979630632599(-4)	500	-3.144737390980985(-3)	6.592900811132393(-6)
51	3.173325912716976(-2)	6.711979630715649(-4)	501	3.144737390980562(-3)	6.592900676535961(-6)
52	9.519977738150902(-2)	6.711979630604550(-4)	502	9.434212172942109(-3)	6.592901619513896(-6)
⋮	⋮	⋮	⋮	⋮	⋮
100	3.14159265358979	98632429463999.1	1000	3.14159265358979	5550622952753.5

Tab. 1: Relative error for $N = 100$ and $N = 1000$.

2 Test on the fundamental of the zero range potential

We now want to test the accuracy on a more difficult function, that is

$$\psi_0(x) = \kappa^{1/2} e^{-\kappa|x|}. \quad (7)$$

The derivative of this function will then be

$$\psi'_0(x) = \begin{cases} \kappa^{3/2} e^{-\kappa x} & \text{if } x \geq 0, \\ -\kappa^{3/2} e^{-\kappa x} & \text{if } x \leq 0. \end{cases} \quad (8)$$

We want to know how well Eq. (4) approximates Eq. (8). As can be seen in Fig. (??), the approximation is not as good as in the case of differentiable functions. Tab (2) and Tab (3) show the evolution of the relative error around $x = 0$ for space grids.

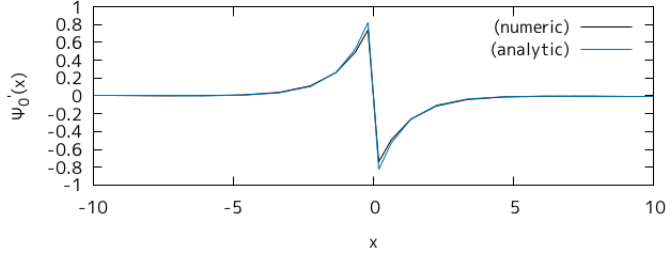


Figure 2: Relative error to the reference for Eq. (4) for a non differentiable function, nmax = 1899

i	x_i	$\psi'_0(x)$ [Eq.(4)]	$\psi'_0(x)$ [Eq. (8)]	σ_{rel}
945	-3.36223597709689	4.003203568686754(-2)	3.465767846769117(-2)	0.155069740871031
946	-2.24631739047944	0.112996198349178	0.105788084319555	6.813729614244640(-2)
947	-1.33122021910050	0.262296156170054	0.264154737908558	-7.035958367505338(-3)
948	-0.640655011873346	0.492411254637094	0.526947154334799	-6.553958857848256(-2)
949	-0.192562951060858	0.738389362290233	0.824842391865603	-0.104811574220662
950	0.000000000000000	0.000000000000000	0.000000000000000	NaN
951	0.192562951060858	-0.738389362290233	-0.824842391865603	-0.104811574220662
952	0.640655011873346	-0.492411254637094	-0.526947154334799	-6.553958857848256(-2)
953	1.33122021910050	-0.262296156170054	-0.264154737908558	-7.035958367505338(-3)
954	2.24631739047944	-0.112996198349178	-0.105788084319555	6.813729614244640(-2)
955	3.36223597709689	-4.003203568686754(-2)	-3.465767846769117(-2)	0.155069740871031

Tab. 2: Relative error between analytical formula and numerical approximation of the derivative of $\varphi'_0(x)$. nbas = 20, nelm = 100 and nmax = 1899.

i	x_i	$\psi'_0(x)$ [Eq.(4)]	$\psi'_0(x)$ [Eq. (8)]	σ_{rel}
3895	-0.428483301548953	0.644931524794986	0.651496469445254	-1.007671562035826(-2)
3896	-0.281801175281992	0.745407386634963	0.754423667695403	-1.195121712973636(-2)
3897	-0.164949617560057	0.836510512443705	0.847936424117700	-1.347496268471325(-2)
3898	-7.866859641248336(-2)	0.910832210644854	0.924346205339547	-1.462005752458130(-2)
3899	-2.351035377825550(-2)	0.961677188998991	0.976763861420060	-1.544556777431891(-2)
3900	0.000000000000000	0.000000000000000	0.000000000000000	NaN
3901	2.351035377825550(-2)	-0.961677188998991	-0.976763861420060	-1.544556777431891(-2)
3902	7.866859641248336(-2)	-0.910832210644854	-0.924346205339547	-1.462005752458130(-2)
3903	0.164949617560057	-0.836510512443705	-0.847936424117700	-1.347496268471325(-2)
3904	0.281801175281992	-0.745407386634963	-0.754423667695403	-1.195121712973636(-2)
3905	0.428483301548953	-0.644931524794986	-0.651496469445254	-1.007671562035826(-2)

Tab. 3: Relative error between analytical formula and numerical approximation of the derivative of $\varphi'_0(x)$. nbas = 40, nelm = 200 and nmax = 7799.

3 Accuracy of Eq. (4) for evaluating the derivative of $\psi(x, t)$

The final test we want to perform is the comparison of the formula

$$Q^{(c)}(\omega) = |p(\omega)|^2; \quad \text{Eq.(21)} \quad (9)$$

with $\hat{p} = -id/dx$, and $p(t)$ and $p(\omega)$ evaluated according to Eq. (2) and (3), to Eq. (1). Moreover, we also want to compare it to the following formula

$$Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2, \quad \text{Eq.(31)} \quad (10)$$

referred to as Eq. (31).

Eq. (9) is designated by Eq. (21) in a previous document we submitted. The results are compiled in Fig. (3). The two last panels with intensities $F_0 = 0.3$ a.u. show how Eq. (9) and (10) compare to Eq. (1).

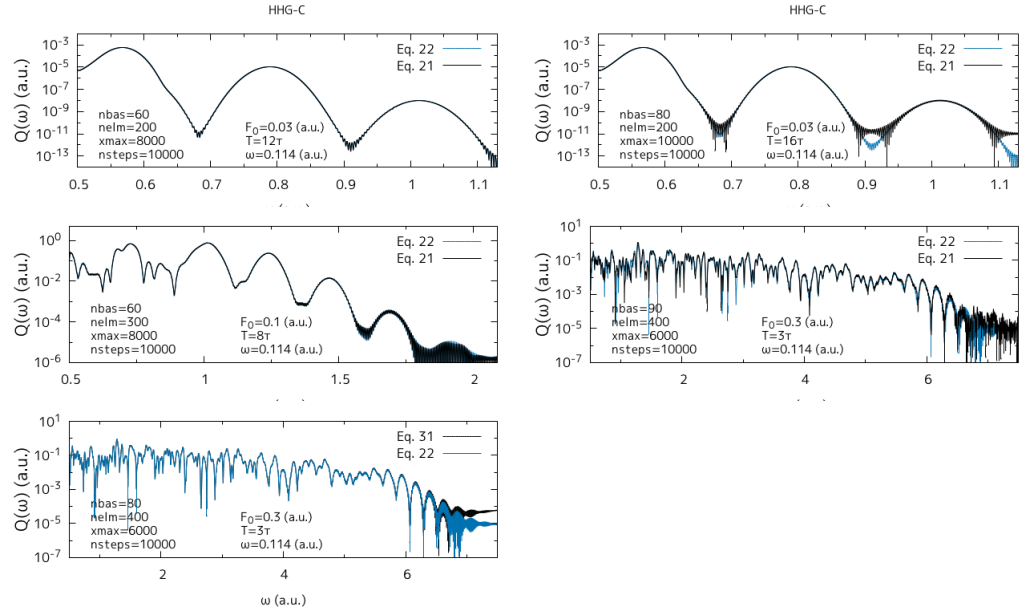


Figure 3: Comparison between Eq. (1) and (9) referred to as Eq. (22) and (21) respectively, for different intensities. The relevant parameters are written on the figures.